Computer Networks Project: FreshMart Enterprise Network Design

Project By:

Fidia Batool 70174381

Ayesha Kashif 70175728

1. Project Overview

This project simulates a scalable enterprise network for "FreshMart," a retail chain with a centralized Headquarters (HQ) and multiple geographically separated Branch stores. The design utilizes a **Hub-and-Spoke** topology to ensure centralized management of services (DNS, Web) while allowing autonomous daily operations at the branches (DHCP, VLANs).

Objectives

- **Scalability:** Use OSPF routing to allow easy addition of new branches.
- **Security:** Use VLANs to separate Point-of-Sale (POS) systems from Management traffic.
- **Efficiency:** Implement VLSM (Variable Length Subnet Masking) to conserve IP addresses.
- **Automation:** Use DHCP for automatic IP assignment and DNS for hostname resolution.

---

2. Network Topology & Hardware

Physical Layout

- **Headquarters (Hub):**
  - **1x Router (2911):** Acts as the Core Router and OSPF backbone.
  - **1x Switch (2960):** Connects the Server Farm.
  - **Servers:** 1x DNS Server, 1x Web Server.
- **Branches (Spokes):**
  - **3x Routers (2911):** Connect the branches to HQ via WAN.
  - **3x Switches (2960):** Manage local VLANs.
  - **End Devices:** PCs for POS (Cashiers) and Management.

Connection Types

- **WAN (Wide Area Network):** Serial DCE/DTE connections simulating leased lines.
- **LAN (Local Area Network):** Gigabit/FastEthernet connections for internal switching.

---

3. IP Addressing Scheme (VLSM)

We utilized a structured IP plan to maximize efficiency.

Location	Device / VLAN	Network Address	Subnet Mask	Gateway IP
HQ	Server Farm	192.168.10.0	255.255.255.0	192.168.10.1
WAN Links	HQ <-> BR1	10.0.0.0	255.255.255.252 (/30)	N/A
	HQ <-> BR2	10.0.0.4	255.255.255.252 (/30)	N/A
Branch 1	VLAN 10 (POS)	192.168.20.0	255.255.255.0	192.168.20.1
	VLAN 20 (Mgmt)	192.168.21.0	255.255.255.0	192.168.21.1
Branch 2	VLAN 10 (POS)	192.168.30.0	255.255.255.0	192.168.30.1
	VLAN 20 (Mgmt)	192.168.31.0	255.255.255.0	192.168.31.1

---

4. Configuration Documentation

A. Headquarters (HQ) Configuration

The HQ acts as the DCE (Service Provider side) for clocking and the central OSPF router.

HQ Router:

```
enable
conf t
hostname HQ_Router

# Internal Gateway for Servers
interface g0/0
 no shutdown
 ip address 192.168.10.1 255.255.255.0

# WAN Link to Branch 1 (DCE side)
interface s0/3/0
 ip address 10.0.0.1 255.255.255.252
 clock rate 128000
 no shutdown

# WAN Link to Branch 2 (DCE side)
interface s0/3/1
 ip address 10.0.0.5 255.255.255.252
 clock rate 128000
 no shutdown

# OSPF Routing (Advertising all networks)
router ospf 1
 network 192.168.10.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0

```

HQ Services:

- **DNS Server:** A-Record created mapping freshmart.com -> 192.168.10.6.
- **Web Server:** HTTP enabled; custom index.html deployed.

B. Branch 1 Configuration

Implements "Router-on-a-Stick" for VLANs and DHCP for local clients.

BR1 Switch:

```
# VLAN Creation
vlan 10
 name POS
vlan 20
 name MGMT

# Port Assignment
interface fa0/2
 switchport mode access
```

```
 switchport access vlan 10
interface fa0/3
 switchport mode access
 switchport access vlan 20

# Trunk Link to Router
interface g0/1
 switchport trunk

```

BR1 Router:

```
# WAN Interface
interface s0/3/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown

# Sub-Interfaces (Gateway for VLANs)
interface g0/0
 no shutdown
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.20.1 255.255.255.0
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.21.1 255.255.255.0

# DHCP Configuration
ip dhcp pool BR1_POS
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.10.5

# OSPF Routing
router ospf 1
 network 192.168.20.0 0.0.1.255 area 0
 network 10.0.0.0 0.0.0.3 area 0

```

C. Branch 2 Configuration

Identical logic to Branch 1, but with unique subnets to avoid conflicts.

BR2 Router:

```
# WAN Interface
interface s0/3/0
 ip address 10.0.0.6 255.255.255.252
 no shutdown

# Sub-Interfaces
interface g0/0
 no shutdown
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.30.1 255.255.255.0

# DHCP Configuration
ip dhcp pool BR2_POS
```

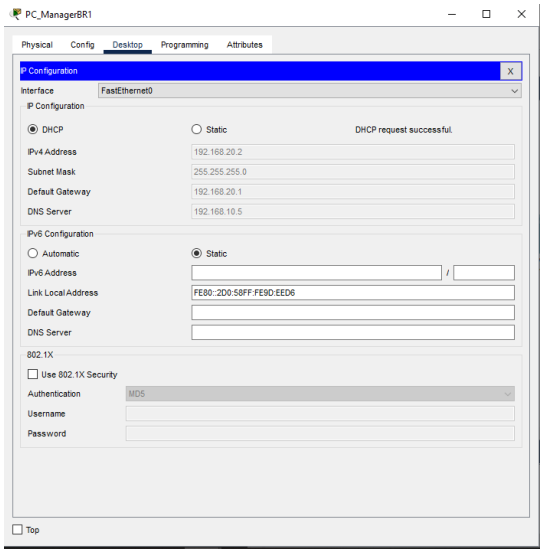
```
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.10.5

# OSPF Routing
router ospf 1
 network 192.168.30.0 0.0.1.255 area 0
 network 10.0.0.4 0.0.0.3 area 0

```

---

5. Testing & Verification

To confirm the network meets all requirements, the following tests were performed:

1. **DHCP Test:**
   - **Action:** PCs were set to "DHCP" mode.
   - **Result:** All PCs successfully received IP addresses, Subnet Masks, Gateways, and the DNS Server IP (192.168.10.5).

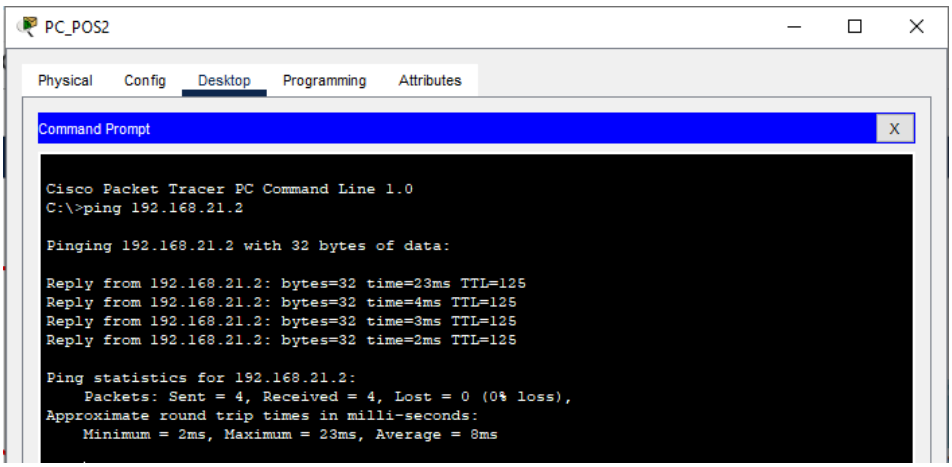


2. **Inter-VLAN Routing Test:**
   - **Action:** Ping from VLAN 10 (POS) to VLAN 20 (Mgmt).
   - **Result:** Successful. Router-on-a-stick is functioning.

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. WAN Connectivity (OSPF) Test:

- **Action:** Ping from Branch 1 PC to Branch 2 PC.
- **Result:** Successful. OSPF is correctly routing traffic via HQ.



4. Web Service Test:

- **Action:** Opened Web Browser on Branch PC and typed freshmart.com.
- **Result:** The "FreshMart Corporate Portal" page loaded successfully.

```
C:\>nslookup freshmart.com

Server:  [192.168.10.5]
Address:  192.168.10.5

Non-authoritative answer:
Name:    freshmart.com
Address:  192.168.10.6
```

---

6. Troubleshooting Log

During implementation, the following issues were encountered and resolved:

- **Issue:** Serial Interfaces showed "Red Arrows" (Down/Down).
  - **Solution:** Applied the clock rate 128000 command on the DCE (HQ) side and no shutdown on both interfaces.
- **Issue:** PCs received APIPA addresses (169.254.x.x).
  - **Solution:** Identified a mismatch in Trunk configuration. Executed switchport mode trunk on the Branch Switch uplink and verified encapsulation dot1Q on the Router sub-interfaces.
- **Issue:** Web page failed to load via hostname.
  - **Solution:** Added the dns-server 192.168.10.5 command to the Branch DHCP pools, ensuring clients knew where to resolve the address.

---

7. Design Justification & Technical Rationale

In designing the FreshMart Enterprise Network, specific technologies were chosen to meet modern industry standards for security, scalability, and operational efficiency. Below is the justification for each implemented concept.

1. Topology: Hub-and-Spoke Architecture

- **Implementation:** The Headquarters (HQ) acts as the central hub, with all Branch offices (Spokes) connecting directly to it via WAN links.
- **Justification:** This topology provides **centralized management**. All traffic flows through the HQ, allowing the IT team to strictly monitor data, manage security policies, and host centralized services (like the Web and DNS servers) in one secure location. It is also cost-effective, as branches do not need expensive links to connect to every other branch, only to the HQ.

2. Variable Length Subnet Masking (VLSM)

- **Implementation:** We used /24 subnets (254 IPs) for LANs and /30 subnets (2 IPs) for WAN links.
- **Justification:** Standard classful addressing wastes IP addresses. By using a /30 mask for the WAN links, we use exactly the 2 addresses needed (one for each

end of the cable). This practice is critical in real-world networks to **conserve public IP address space** and allows for efficient route summarization.

3. VLANs (Virtual Local Area Networks)

- **Implementation:** Created **VLAN 10 (POS)** and **VLAN 20 (Management)** to separate traffic at the switch level.
- **Justification: Security and Compliance.** In a retail environment, Point-of-Sale (POS) systems handle sensitive credit card data. Keeping them on a separate VLAN ensures that a compromised Management PC cannot sniff or access payment traffic. This aligns with industry standards like **PCI-DSS**.

4. Inter-VLAN Routing (Router-on-a-Stick)

- **Implementation:** Configured Sub-interfaces (g0/0.10, g0/0.20) on the Branch Routers to route traffic between VLANs using a single physical cable.
- **Justification: Cost Efficiency.** Without this, we would need a separate physical router port for every VLAN, which is expensive and unscalable. "Router-on-a-Stick" allows a single router interface to handle traffic for multiple departments effectively.

5. OSPF (Open Shortest Path First)

- **Implementation:** Enabled OSPF Area 0 on all routers to exchange route information dynamically.
- **Justification: Scalability.** Unlike Static Routing, which requires manual updates every time the network changes, OSPF allows the HQ router to automatically "learn" about new branches. If FreshMart opens 50 new stores, OSPF will automatically route traffic to them without requiring manual reconfiguration of the existing core network.

6. DHCP (Dynamic Host Configuration Protocol)

- **Implementation:** Configured Branch Routers to automatically assign IPs, Gateways, and DNS addresses to local PCs.
- **Justification: Operational Efficiency.** In a real store, employees are not technical engineers. DHCP ensures that when a replacement cash register or laptop is plugged in, it works immediately without requiring an IT technician to visit the store to manually type in IP addresses.

7. DNS (Domain Name System)

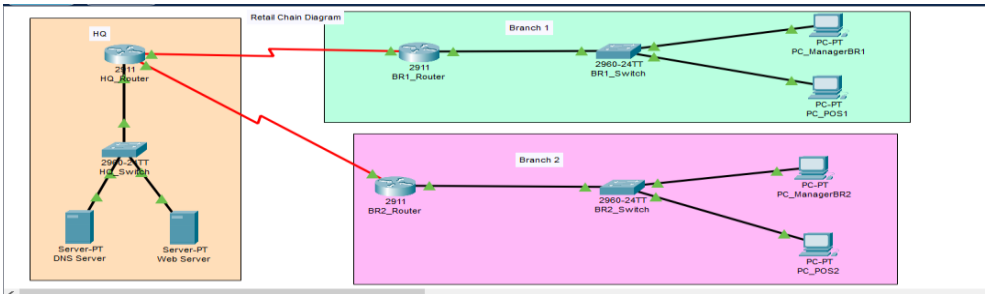
- **Implementation:** A central server maps freshmart.com to 192.168.10.6.
- **Justification: User Experience (UX).** Humans remember names, not numbers. DNS ensures that employees can access internal resources using intuitive URLs. Centralizing this at HQ ensures that if the Web Server IP changes, we only need to update one record at HQ, rather than updating every PC in every branch.

8. Serial WAN Connections

- **Implementation:** Connected HQ to Branches using Serial DCE/DTE cables with a set clock rate.
- **Justification: Simulation of Reality.** This simulates a "Leased Line" connection from an Internet Service Provider (ISP). In the real world, businesses pay for dedicated lines for reliability. Configuring the **Clock Rate** on the HQ side correctly simulates the synchronization required for synchronous WAN data transmission.

---

9. Conclusion

The FreshMart network design successfully demonstrates a modern, scalable enterprise architecture. By separating logical functions (VLANs), automating management (DHCP/DNS), and utilizing dynamic routing (OSPF), the network is robust, secure, and ready for future expansion.

Complete Network Diagram